

MINI SOC LAB

Network Traffic Monitoring & Threat Detection

Tools: Wireshark | Nmap | Windows Command Prompt

Environment: Personal Windows Laptop | Wi-Fi Network

Host IP: 192.168.100.230

April 2026

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1. Introduction

Network monitoring is one of the core skills I wanted to develop as part of my cybersecurity learning. For this project, I set up a small SOC-style lab on my personal laptop to capture and analyse real network traffic using Wireshark and Nmap. The goal was straightforward — observe what normal traffic looks like, then compare it against traffic that shows signs of suspicious behaviour.

I structured the lab around two phases. In the first, I generated traffic through browsing and ping commands. In the second, I used packet filters to dig into TCP SYN behaviour and understand what scan-like activity looks like in a capture. Working through both phases gave me hands-on experience with packet analysis, protocol identification, and the basics of threat detection.

2. Objective

The objective of this project was to build a mini-SOC lab that captures and analyses network traffic in order to identify the difference between normal network behaviour and suspicious scan-like activity.

3. Tools Used

- Wireshark — packet capture and protocol analysis
- Nmap — network scanning and security testing
- Windows Command Prompt — traffic generation (ping, ipconfig)
- Wi-Fi network connection — live capture interface
- Personal Windows laptop — host environment

4. Lab Environment

The lab was conducted on a Windows laptop connected to a Wi-Fi network. Wireshark was used to capture live packets from the wireless interface, while Command Prompt was used to generate traffic such as ping requests. Nmap was installed to support security testing and scan-related analysis.

Host IP observed during captures: 192.168.100.230

5. Methodology

5.1 Normal Traffic Capture

To start, I opened Wireshark and began a live capture on my Wi-Fi interface. I browsed a few websites to generate realistic traffic, then ran a ping google.com command from Command Prompt to add some ICMP packets to the mix.

The purpose of this stage was to identify protocols commonly seen in everyday network usage, including DNS, ICMP, and TLS.

5.2 Suspicious Traffic Analysis

```
| tcp.flags.syn == 1 and tcp.flags.ack == 0
```

This filter highlights initial TCP connection attempts, which are often associated with reconnaissance or scan activity. The resulting traffic was reviewed to identify repeated SYN packets and unusual connection patterns.

6. Analysis and Findings

6.1 DNS Traffic Analysis

The DNS filter in Wireshark displayed multiple domain name resolution requests generated during browsing activity. These queries included requests to external services and websites such as bing.com, msn.com, and scorecardresearch.com, showing the expected behaviour of a host translating domain names into IP addresses.

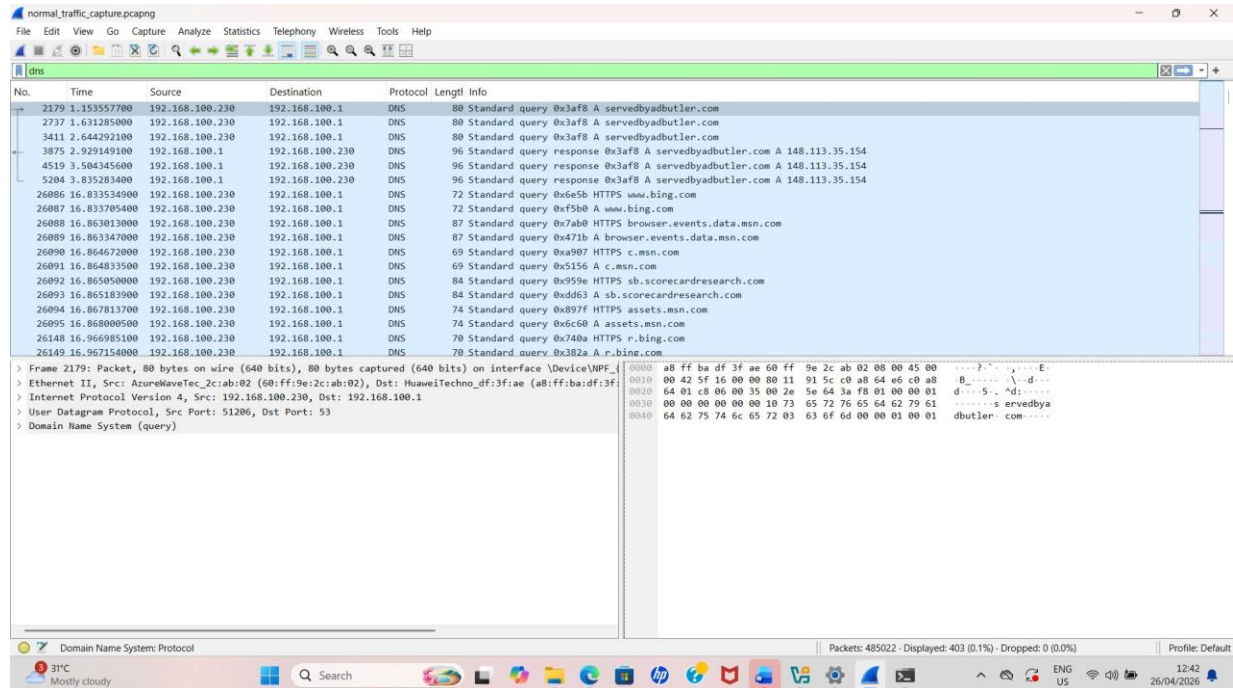


Figure 1: DNS traffic captured in Wireshark showing standard query and response activity during normal browsing.

The observed DNS packets indicate legitimate, expected communication. DNS is a normal part of browsing activity, and the presence of standard queries and responses confirms normal host behaviour.

6.2 ICMP Traffic Analysis

ICMP traffic was generated using the ping google.com command. The Wireshark ICMP filter successfully displayed echo requests and replies between the host and the destination (142.250.202.142 — Google). The ping completed with 0% packet loss and an average round-trip time of 18ms, confirming healthy connectivity.

```

Microsoft Windows [Version 10.0.26100.1301]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Hajer>ping google.com

Pinging google.com [142.250.202.142] with 32 bytes of data:
Reply from 142.250.202.142: bytes=32 time=18ms TTL=117
Reply from 142.250.202.142: bytes=32 time=17ms TTL=117
Reply from 142.250.202.142: bytes=32 time=17ms TTL=117
Reply from 142.250.202.142: bytes=32 time=20ms TTL=117

Ping statistics for 142.250.202.142:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 17ms, Maximum = 20ms, Average = 18ms

C:\Users\Hajer>

```

Figure 2: Ping command executed in Command Prompt to generate ICMP traffic (google.com resolved to 142.250.202.142).

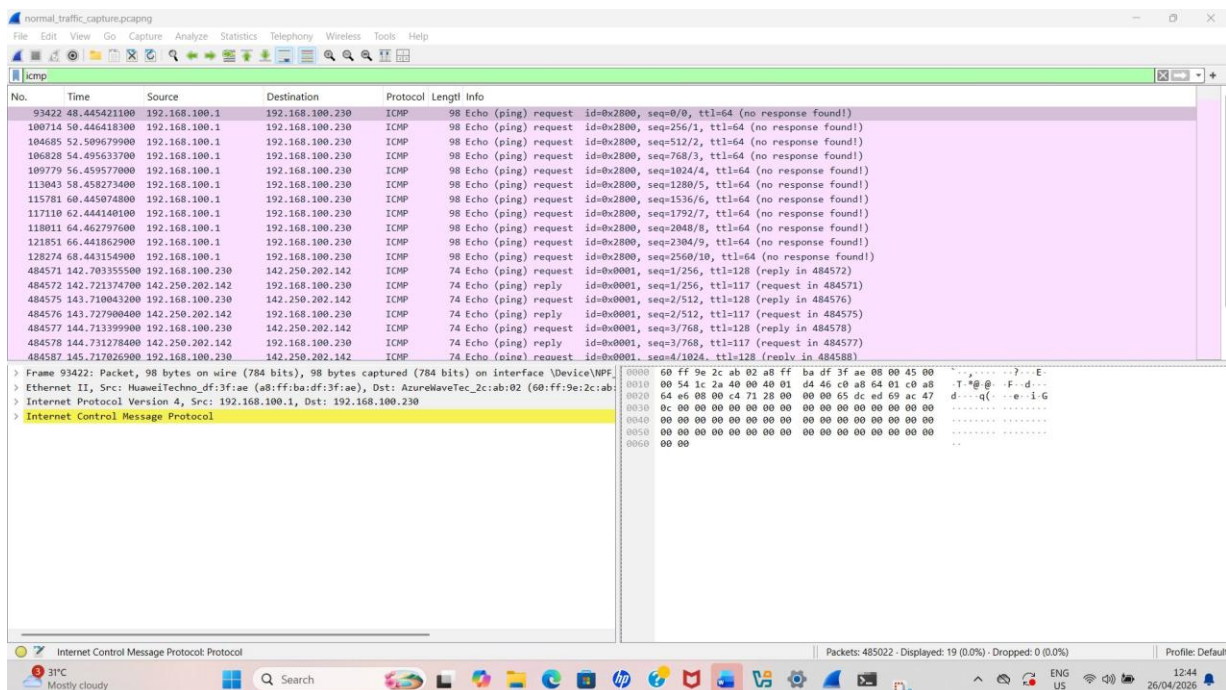


Figure 3: ICMP packets captured in Wireshark showing echo request and echo reply messages.

This confirms that the environment successfully captured diagnostic traffic. ICMP requests and replies are commonly used for connectivity testing and are consistent with normal administrative or troubleshooting behaviour.

6.3 TLS Traffic Analysis

TLS traffic was captured during normal browsing and filtered in Wireshark. The packets showed encrypted application data exchanged over port 443, which is standard for secure HTTPS communication. Both TLSv1.2 and TLSv1.3 sessions were observed.

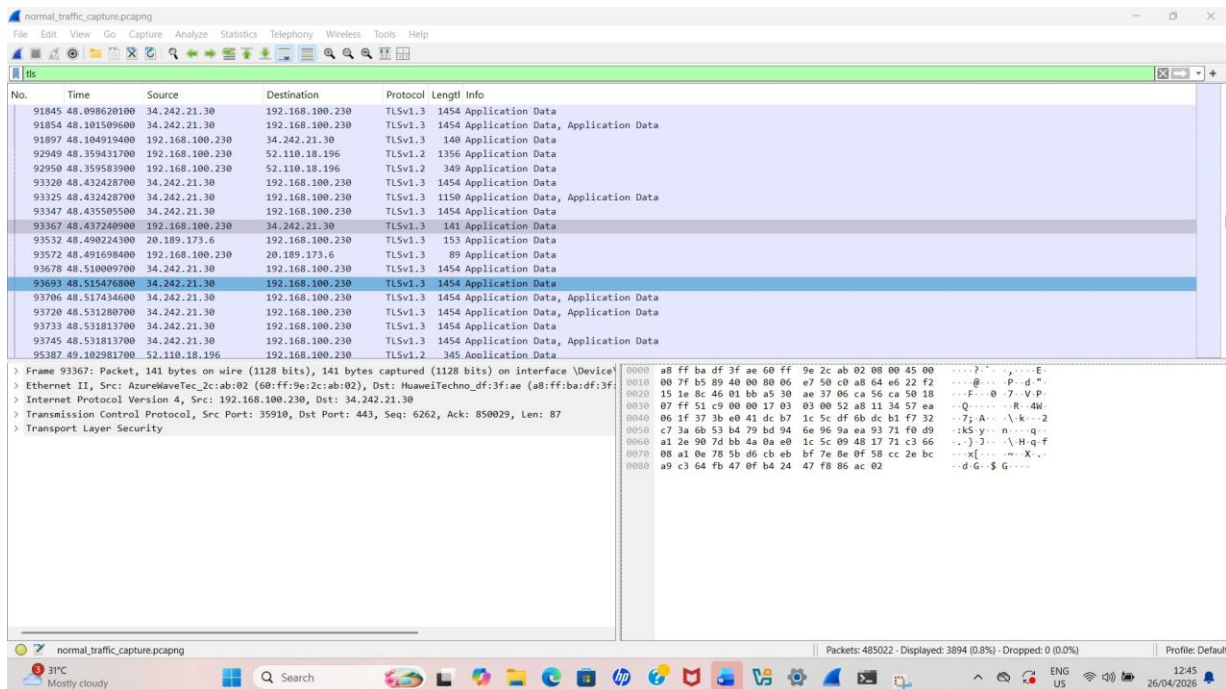


Figure 4: TLS traffic captured in Wireshark showing encrypted web communication (TLSv1.2 and TLSv1.3).

The appearance of TLSv1.2 and TLSv1.3 packets confirms that the user's browsing activity included encrypted sessions with external servers. This is typical for modern web traffic and represents normal network behaviour.

6.4 Live Network Monitoring

A live capture session showed mixed traffic types before filtering, including QUIC, TLS, TCP, MDNS, and multicast-related packets. This reflects how real network traffic appears in a raw capture before analysis is narrowed to a specific protocol or suspicious behaviour.

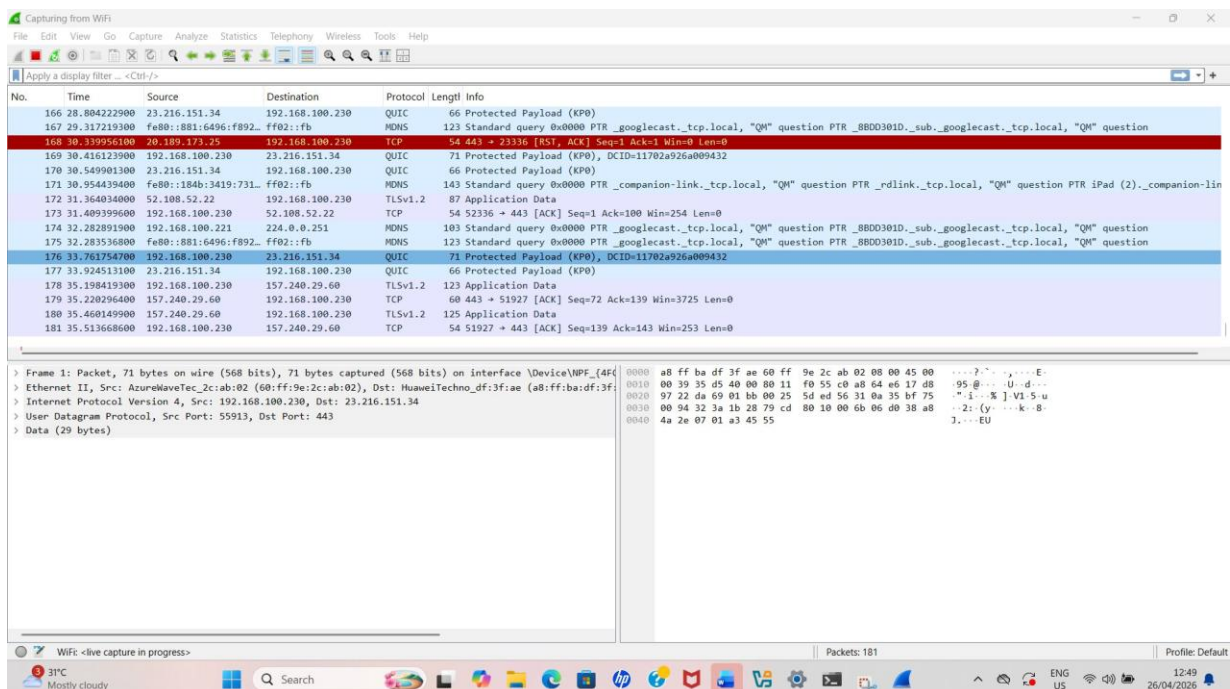


Figure 5: Live Wireshark capture during network monitoring before applying focused filters.

This stage demonstrated the importance of filtering during SOC analysis. Raw traffic can contain many unrelated packets, so analysts must use protocol filters to isolate meaningful behaviour.

6.5 TCP SYN Scan-Like Activity

The most important suspicious indicator in this project was observed using the TCP SYN filter :

```
tcp.flags.syn == 1 and tcp.flags.ack == 0
```

The filtered results displayed multiple SYN packets from the host (192.168.100.230) to destination systems, including repeated connection attempts to port 53 (DNS) and port 443 (HTTPS). SYN packets represent the beginning of a TCP handshake, and repeated SYN activity within a short period can indicate scan or reconnaissance behaviour.

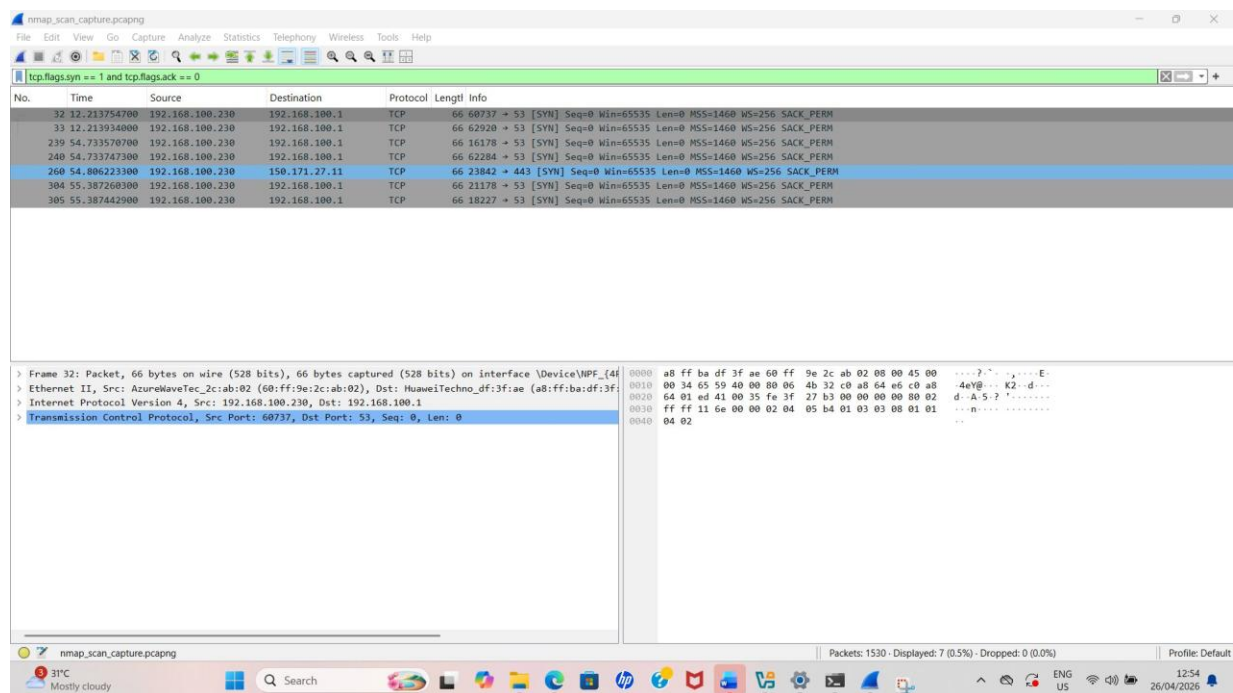


Figure 6: TCP SYN packets captured in Wireshark indicating repeated connection attempts consistent with scan-like behaviour.

The additional ping test to the host IP (192.168.100.230) confirmed sub-millisecond round-trip times, verifying the host was active and responsive on the local network.

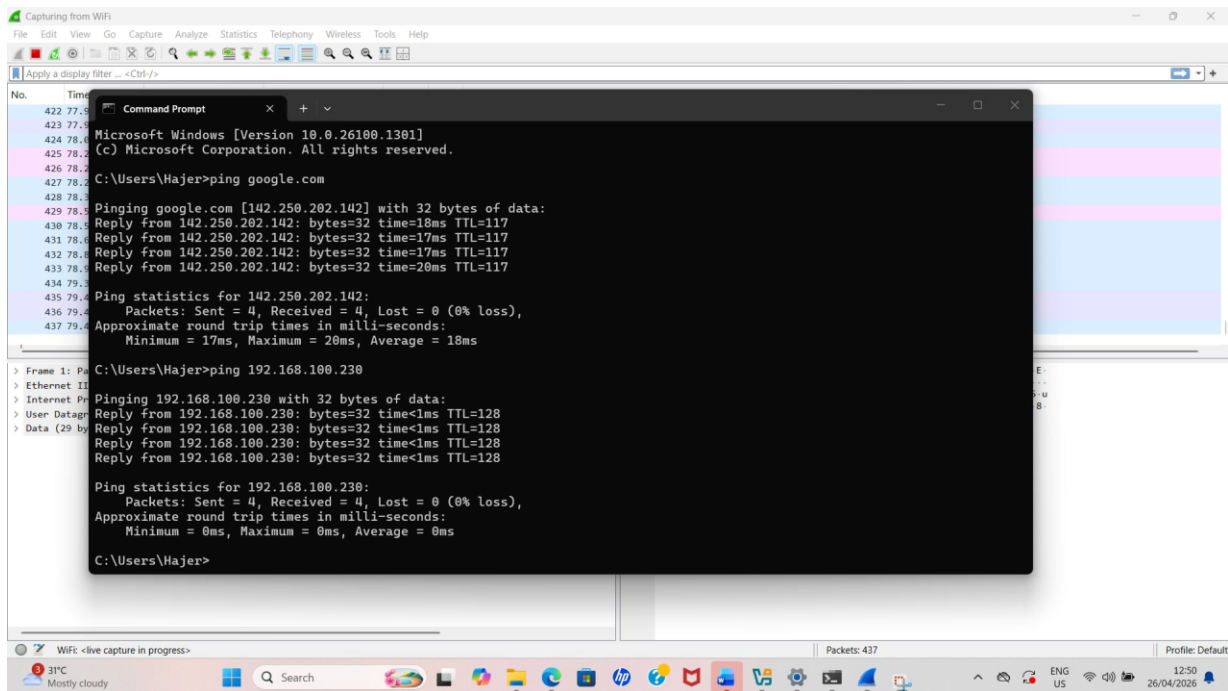


Figure 7: Ping to host IP 192.168.100.230 confirming local network connectivity with <1ms response time.

Compared with normal browsing traffic, this behaviour appeared more repetitive and targeted. While normal traffic usually includes completed sessions and encrypted exchanges, SYN-focused traffic highlights initial probing activity. In a SOC environment, this kind of pattern may be treated as an indicator of suspicious reconnaissance and should be investigated further.

7. Summary of Key Findings

The table below summarizes the protocols observed and their security classification:

Protocol	Type	Verdict
DNS	Normal	✓ Legitimate
ICMP	Normal	✓ Legitimate
TLS 1.2/1.3	Normal	✓ Legitimate
QUIC/TCP	Normal	✓ Legitimate
TCP SYN only	Suspicious	⚠ Investigate

- Normal traffic included DNS, ICMP, and TLS packets consistent with regular web browsing.
- DNS traffic showed standard domain resolution requests to services like bing.com and msn.com.
- ICMP packets confirmed successful connectivity testing using ping to Google (0% packet loss).
- TLS traffic demonstrated encrypted web communication over HTTPS (TLSv1.2 and TLSv1.3).

- Live captures contained mixed packet types (QUIC, MDNS, TCP), showing the need for targeted filtering.
- TCP SYN packets revealed repeated connection attempts to ports 53 and 443, consistent with scan-like activity.
- Suspicious traffic patterns were more repetitive and narrowly focused than normal user activity.

8. Security Interpretation

What this lab reinforced for me is that suspicious traffic doesn't always look dramatic. The SYN packets I flagged are the same protocol used in every legitimate TCP connection — the difference is the pattern. Repeated, unanswered SYN requests across multiple ports is what makes it worth investigating, not the protocol itself.

It also made me think more carefully about context. A single SYN packet means nothing. Ten of them in two seconds, all from the same source, targeting different ports — that's a different story. Learning to read that difference is, I think, one of the more practical skills that comes out of doing this kind of hands-on analysis rather than just studying theory.

9. Recommendations

- Monitor repeated TCP SYN activity as a potential indicator of reconnaissance.
- Use Wireshark filters to isolate suspicious behaviour during packet analysis.
- Investigate hosts that generate frequent connection attempts to multiple services or ports.
- Combine packet analysis with IDS/IPS tools for stronger scan detection.
- Maintain baseline knowledge of normal traffic patterns to improve anomaly detection.

10. Conclusion

This lab gave me something I don't think I could have gotten just from reading — a real sense of what network traffic looks and feels like at the packet level. Setting everything up myself, running the captures, and working through the filters made the concepts stick in a way that theory alone doesn't.

The project strengthened foundational skills in packet capture, protocol analysis, filtering, and basic threat detection. It also showed how network traffic analysis can support cybersecurity monitoring and help identify indicators of suspicious behaviour in real environments.

11. Figure List

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- Figure 2: Ping command executed in Command Prompt to generate ICMP traffic.
- Figure 3: ICMP packets captured in Wireshark showing echo request and echo reply messages.
- Figure 4: TLS traffic captured in Wireshark showing encrypted web communication.
- Figure 5: Live Wireshark capture during network monitoring before applying focused filters.

- Figure 6: TCP SYN packets captured in Wireshark indicating repeated connection attempts consistent with scan-like behaviour.
- Figure 7: Ping to host IP 192.168.100.230 confirming local network connectivity.

12. Appendix: Filters Used

```
dns
icmp
tls
tcp.flags.syn == 1 and tcp.flags.ack == 0
ip.addr == 192.168.100.230
tcp
```